

Data visualization guide

How McKesson Workforce Innovation turns workforce data into charts that lead to decisions.

Version 1.0 (review draft) October 2026 All example data is synthetic

OVERVIEW

Start here: the quick path

You don't need all of it. These pages cover most of what goes into a deck; the rest is reference for when you build a particular chart.

On screen, each chart family shows its anatomy and best practice first; variants and build steps open with **More**. The short **Quick guide** PDF has just the pages below.

1 · READ ONCE (20 MINUTES)

- **Start with the question** 7
- **Anatomy of a chart** 9
- **Write the headline last** 10
- **Grey first, then color the point** 11
- **Honest scales** 15
- **People data is personal** 16

2 · BEFORE EVERY DECK

- **The McKesson slide** 19
- **Clarity first** 20
- **Two charts on a slide** 21
- **Flat, not rounded** 22
- **Chart chooser** 5
- **The common traps** 6

3 · WHEN YOU BUILD

- **Bar chart, the default** 31
- **Line chart, over time** 47
- **Set up your tools once** 112
- **Cheat sheet** 110
- **Check before you send** 113

Any other chart: find it in the chooser, read its anatomy and best practice, and open **More** for the build steps.

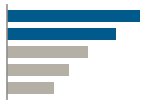
CHART CHOOSER

Chart chooser

Pick the chart from the question, not from habit.

1. Write the question the chart answers in one sentence.
2. Find the question type and start with its first choice.
3. Check the traps on the next page.
4. Follow the chart's page to build it.

Every chart name links to its page in this guide. **Avoid** lists the charts that look right for the question but mislead or slow the reader.



Magnitude

How big? Which is bigger, by how much?

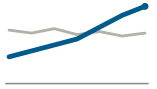
START WITH	**Sorted bar**
ALSO	Dot plot for values far from zero; paired bars
AVOID	Rainbow, 3D or cut-off bars



Part-to-whole

What is it made of? How does a total break down?

START WITH	**100% stacked bar**
ALSO	Waterfall between totals; treemap for many parts
AVOID	Pies over 5 slices



Change over time

How is it moving? When did it change?

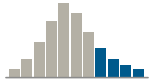
START WITH	**Line**, focus series in blue
ALSO	Stacked area; small multiples; bubble timeline
AVOID	Ten colored lines



Ranking

Who's on top? Who moved?

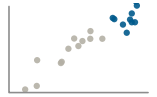
START WITH	**Sorted bar**; **slope** for change
ALSO	Dumbbell for us vs. them; heat table
AVOID	A color for every line



Distribution

How spread out? What's typical?

START WITH	**Strip plot** or **histogram**
ALSO	Box plot to compare many groups
AVOID	An average with no spread



Correlation

Do two measures move together?

START WITH	**Scatter** or bubble scatter
ALSO	Heatmap: two categories, one measure
AVOID	Dual axes



Relationships and flows

How do things connect or move?

START WITH	**Sankey**
ALSO	Parallel coordinates; Venn, 3 sets max
AVOID	A color per stage



Geo-spatial

Where? Does place matter?

START WITH	**Choropleth** for rates
ALSO	Bubble map for counts; world map; zoom-in
AVOID	Counts on a choropleth



Single number

What one figure should they remember?

START WITH	**Big number** with a comparison
ALSO	Progress bar or ring for a target
AVOID	Speedometers

CHART CHOOSER

Before you build: the common traps

The right chart type can still go wrong. If you are about to do one of these, use the fix instead.

The page numbers refer to this guide's PDF; on screen, each fix links to its page.

ABOUT TO	INSTEAD	
Cut the bar axis so small differences show	Bars from zero; a dot plot for a narrow range	p. 33
Give each bar its own color	Grey bars, the one that matters in blue	p. 33
Put two units on two axes	Two aligned panels, or index both to 100	p. 76
Draw ten lines in ten colors	All grey, one or two in blue; or small multiples	p. 49
Make a pie with eight slices	A sorted bar; donuts take five parts at most	p. 45
Stack eight segments in a bar	Five at most, the story segment blue on the baseline	p. 38
Color a waterfall green up, red down	Favorable green, unfavorable orange; or grey with one blue	p. 41
Map raw counts by state	Map rates; show counts as bubbles	p. 94
Show a KPI as a speedometer	A big number with a comparison, or a bar to target	p. 101
Color workstreams red and green, and nothing else	Symbol and word with every dot; status colors only for status	p. 107
Paste a 30-row table to make one point	Chart the point; keep the table for the appendix	p. 64
Put four charts on one slide	Two at most, each with its own insight headline	p. 21
Show a team of 7 people	Suppress or combine groups under 10	p. 16

FUNDAMENTALS

Start with the question, not the chart

A chart is finished when a reader can say what it means and what to do about it, without you in the room.

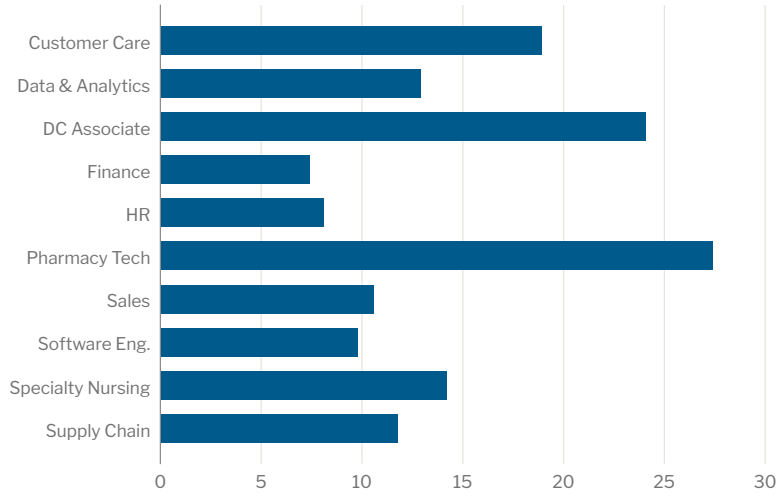
Before you open a tool, write the question in one sentence: "Where is attrition concentrated?", not "show attrition".

Then pick the mode. **Explanatory** charts prove a finding, for decks and reports (the default). **Exploratory** views let readers find answers, for dashboards. Both follow the same rules.

Both charts here are accurate; only one is useful in a meeting.

Attrition by job family

FY26

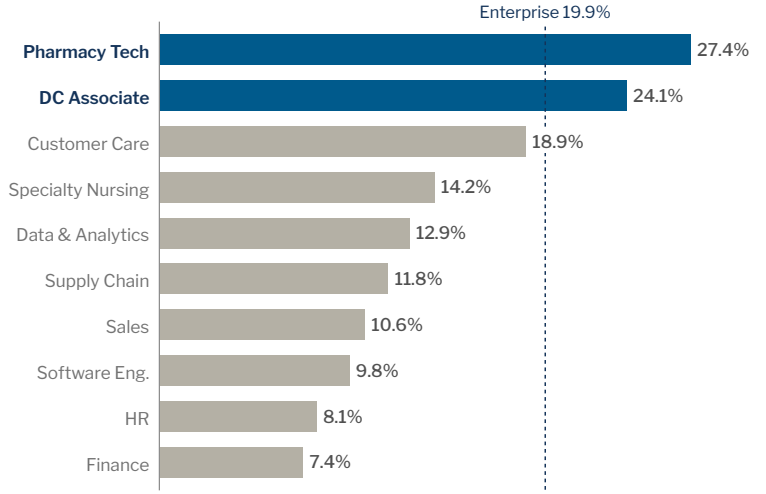


Shown

A label for a title, alphabetical order, every bar equal. The reader has to find the point alone.

Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate

Voluntary attrition by job family, FY26 (%)



Explained

The headline states the finding, the order ranks the families, and color points to the two the headline names.

FUNDAMENTALS

Anatomy of a chart

Every McKesson chart has the same four layers, top to bottom: headline, subtitle, chart, and footnote with source.

The headline and subtitle do different jobs. The headline is editorial: it states the finding. The subtitle is neutral: metric, unit, scope and period.

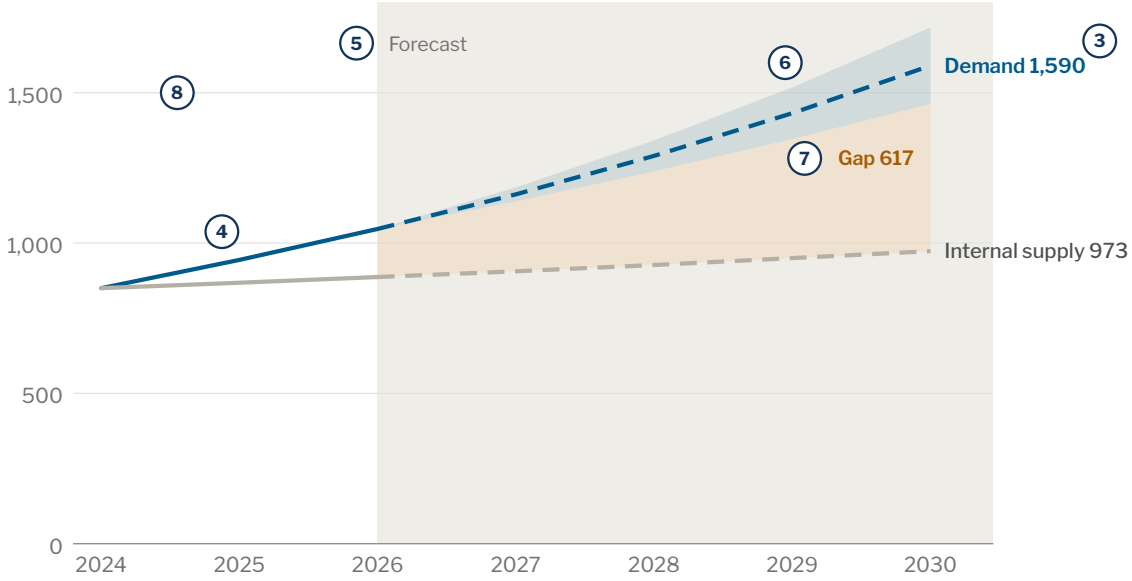
Inside the chart, everything is labelled in place, so the reader never has to match colors to a legend.

All four layers share one left edge. In HTML they are page text; the chart library draws only the data.

1. **Insight headline**
   A full sentence with the number. At most two lines, no final period.
2. **Subtitle**
   Metric, unit, scope and period. No opinions.
3. **Direct labels**
   Series named at the line end, in the series color. No legend.
4. **Focus and context**
   The series the headline is about is blue; the comparison is grey.
5. **Forecast marked**
   A shaded period and dashed lines separate estimates from actuals.
6. **Range, not a point**
   Forecasts show their uncertainty.
7. **The point, annotated**
   The gap the headline quotes, labelled where it is.
8. **Light scaffolding**
   Few, light gridlines; units live in the subtitle.
9. **Footnote and source**
   Definitions, system, as-of date, population.

Internal supply covers only 61% of Data & AI demand by 2030, a gap of about 620 FTE (1)

Data & AI roles: forecast demand vs. internal supply, 2024–2030 (FTE) (2)



Forecast range on demand widens to ±8% by 2030. Internal supply = current staff plus planned reskilling, less expected attrition.

Source: Workday HCM and SWP model (synthetic data for illustration), as of 31 Mar 2026 (9)

FUNDAMENTALS

Write the headline last, from the numbers

The headline is the most-read text on the page. It should be the finding, not the topic.

The formula

Who or what + verb + number + so what.

Before you keep it

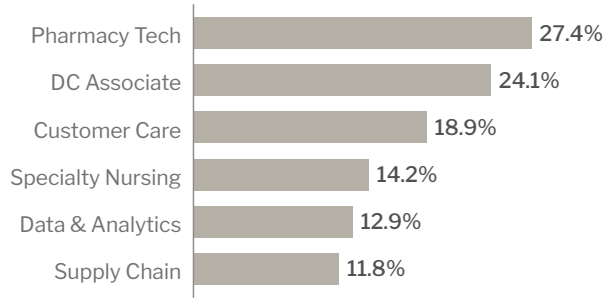
- [ ] Recompute every number, "all", "most" and "n of m" from the data
- [ ] The blue in the chart is what the headline names
- [ ] Two lines at most; no final period

If you can't write a headline, you don't have a finding yet: you have a chart.

LABEL	Attrition by job family · names the topic; says nothing
VAGUE	Attrition varies across job families · true of every chart ever made
OPINION	Attrition is out of control in the frontline · a claim the chart can't support
INSIGHT	**Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate**
INSIGHT + ACTION	**Two frontline families account for 73% of voluntary exits; retention there moves the enterprise rate**

Attrition varies across job families

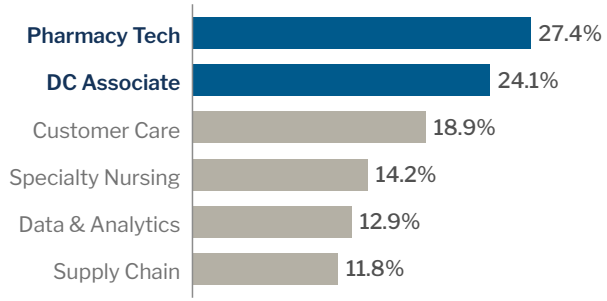
Voluntary attrition by job family, FY26 (%)



Headline and chart both neutral
Nothing tells the reader where to look.

Pharmacy and DC teams lose about 1 in 4 people a year

Voluntary attrition by job family, FY26 (%)



Headline and highlight agree
The two blue bars are exactly what the sentence names.

FUNDAMENTALS

Grey first, then color the point

Color is for meaning, never decoration. Start with every mark in grey, then color only what the headline names.

Highlight at most two items. When everything is highlighted, nothing is.

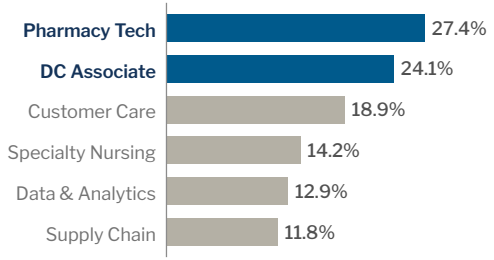
Echo the highlight in the text: bold the matching category label, and set callouts in the color of the mark they describe.

Orange means "worse". Keep it for negatives: below target, shrinking, a shortfall. Never use it as a plain highlight; readers take it as a warning.

PATTERN 1 · DEFAULT

Focus

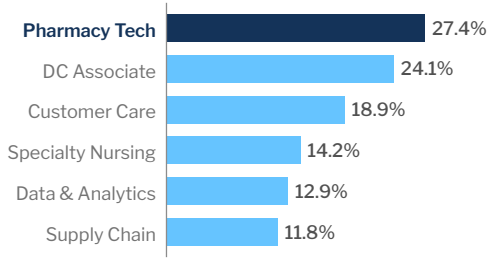
All grey; the items the headline names in blue.



PATTERN 2

Uniform + accent

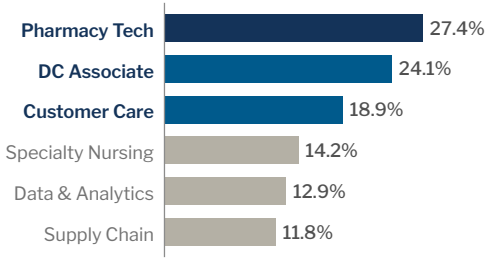
One series in light blue; the single exception in navy.



PATTERN 3

Focus + accent

Grey context, a blue group, one navy exception. Use sparingly.



Color roles

Context · everything the reader should see but not focus on	`#B4B0A5`
Focus · what the headline names	`#005A8C`
Accent · the one exception that must pop (navy, not orange)	`#143359`
Light blue · the series when navy marks the exception	`#66C4FF`
Orange · only when it means worse: below target, shrinking, a shortfall	`#EF8200`
Green · only when it means better, with ▲ or words	`#00AF68`
Total and benchmark · totals, targets, reference lines	`#143359`
Context light · de-emphasized fills in stacks and areas	`#DEDBD1`

FUNDAMENTALS

Honest scales

A chart can be accurate in every number and still mislead. These four rules prevent the common ways.

Bars start at zero. If the differences only show on a narrow range, change the chart, not the axis.

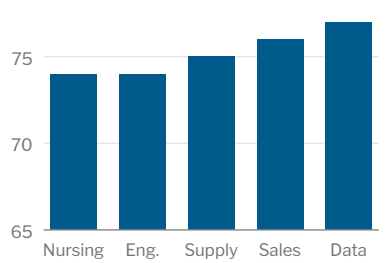
No dual axes. Two scales let you make any two lines cross wherever you like.

Bubble area, not diameter, is proportional to value.

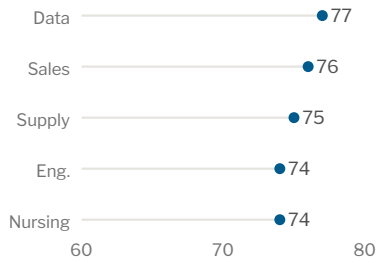
Pies take five slices at most, and sum to 100%.

The one exception: a waterfall bridging two large totals may break its axis, if the break is drawn and footnoted.

Truncated bars

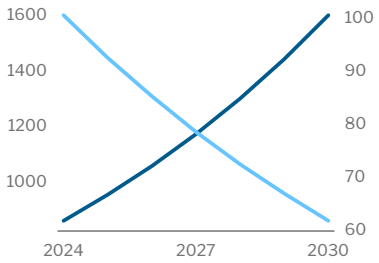


Try to avoid
An axis from 65 turns a 4% difference into a 33% one.

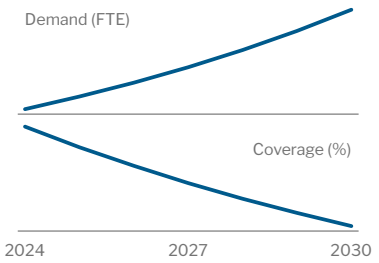


Recommended
A dot plot compares position, so the range can be narrow.

Dual axes

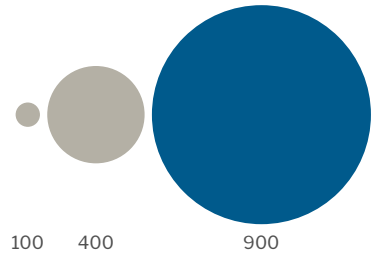


Try to avoid
The crossing point is an artifact of two chosen scales.

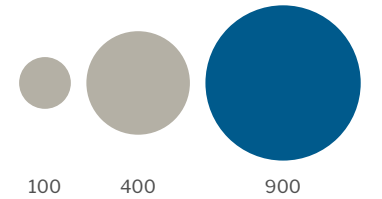


Recommended
Two aligned charts share the time axis.

Bubble size

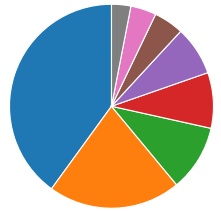


Try to avoid
Sized by diameter, 900 looks 81× bigger than 100.

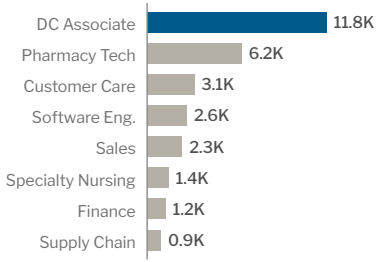


Recommended
Sized by area, 900 is 9× bigger, as it should be.

Too many slices



Try to avoid
Eight similar angles can't be ranked by eye.



Recommended
A sorted bar ranks them instantly.

FUNDAMENTALS

People data is personal

Workforce charts describe colleagues, and carry extra obligations.

Minimum group size

No group under **10 people**: not in bars, labels, tooltips, filters or drill-downs. Merge into "Other" and footnote it.

No re-identification

Small team × tenure × location can single out one person, even when each cut is large.

Protected attributes and forecasts

Protected attributes only for approved analyses such as pay equity, aggregated. Label estimates and show a range. Frame findings about roles, not individuals.

Engagement by team, one distribution center

Two teams have fewer than 10 respondents: anyone who knows the team can guess who answered what.

Team	Respondents	Engagement
Inbound	46	71
Outbound	38	68
Returns	7	52
Quality	4	88
Site leadership	9	81

Try to avoid
Reporting groups below the minimum, even in a table or tooltip.

Engagement by team, one distribution center

Small teams combined into one group that clears the threshold.

Team	Respondents	Engagement
Inbound	46	71
Outbound	38	68
Other teams (3 combined)	20	72

Recommended
Merge into "Other" and footnote it: "Teams with fewer than 10 respondents are combined."

In dashboards, enforce it in the data model

A rule in a slide is easy to forget; a rule in the measure can't be skipped. In Power BI: `IF([Headcount] < 10, BLANK(), [Engagement])`. In Python, drop or merge small groups before plotting, and keep individual values out of hover text.

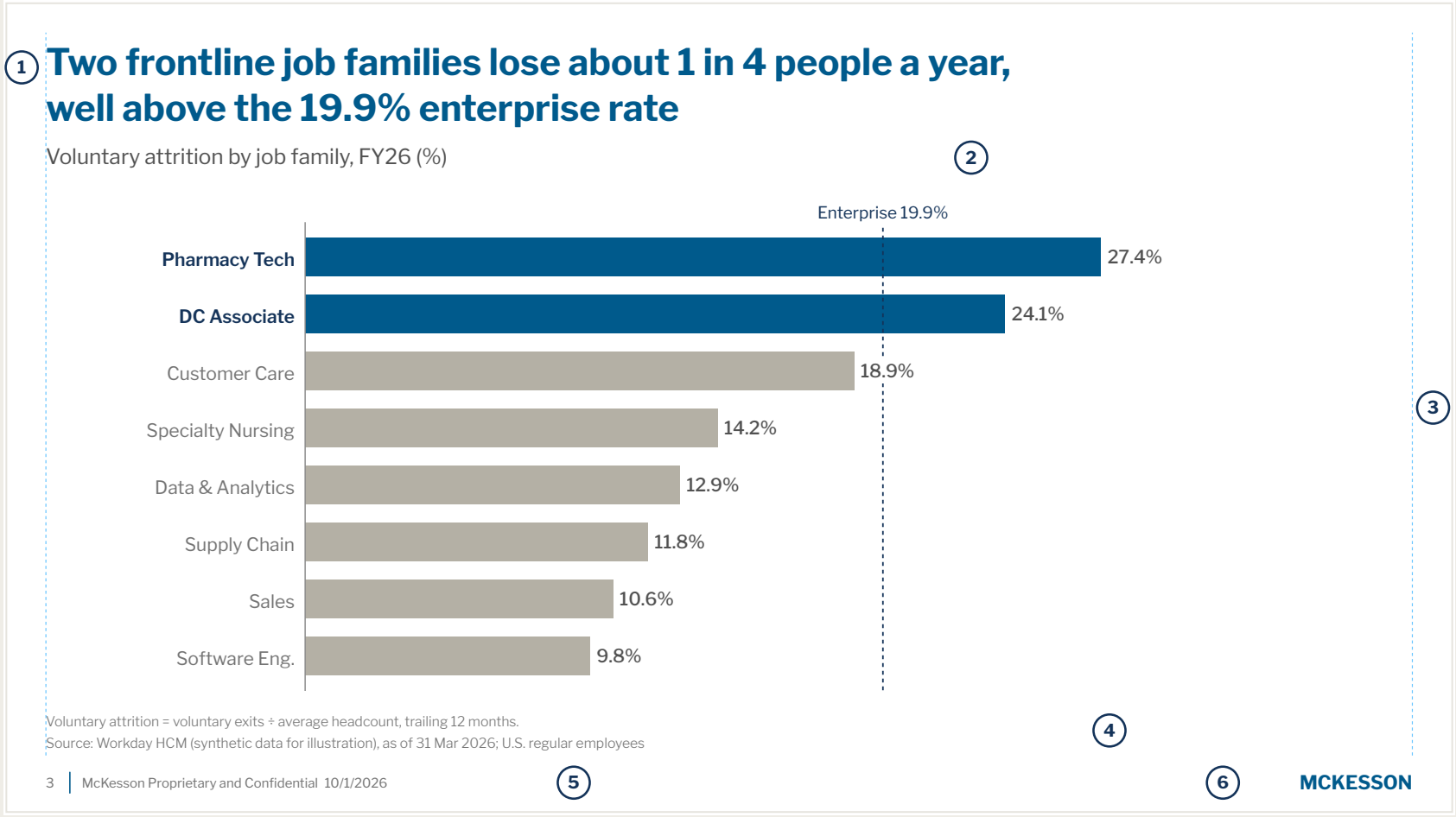
FORMATS

The McKesson slide

Start from the McKesson Standard Template's *Title Only* layout: grid, footer and logo come with it.

1. **Insight title**
   Bold, McKesson Blue, 24–28 pt; two lines at most.
2. **Subtitle**
   Metric, unit, scope, period.
3. **Margins**
   All inside 0.39 in, one left edge.
4. **Source**
   Ends just above the footer.
5. **Footer**
   Page, confidentiality, date: leave on.
6. **Logo**
   From the master; never move it.

The orange underscore is for covers and section headers only: on chart slides, orange means "worse".



FORMATS

Presentations: clarity first

A slide is on screen for less than a minute. There is no time to explore it.

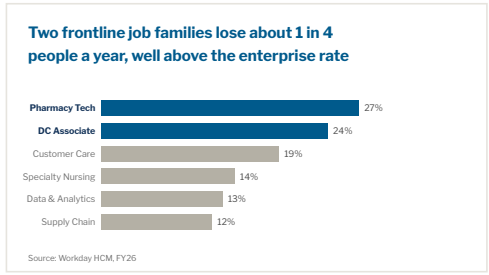
- With one chart, the slide title **is** its headline (Bold)
- One message per chart; at most two charts per slide, never a chart wall
- Short text beside the chart, or none
- Native, editable charts from the starter deck, not pictures of charts
- In a dimmed room, use the dark slide: navy background, light-blue focus

Chart wall



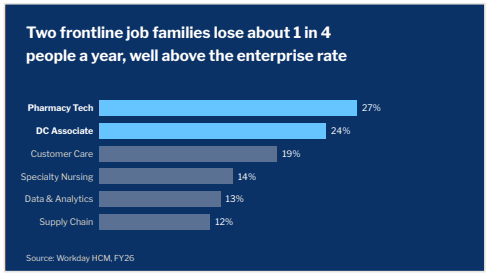
Try to avoid
A label title, four charts and a column of bullets.

One message



Recommended
Headline as title, one chart, a source line.

Dimmed room



Recommended
The same slide on navy; focus in light blue.

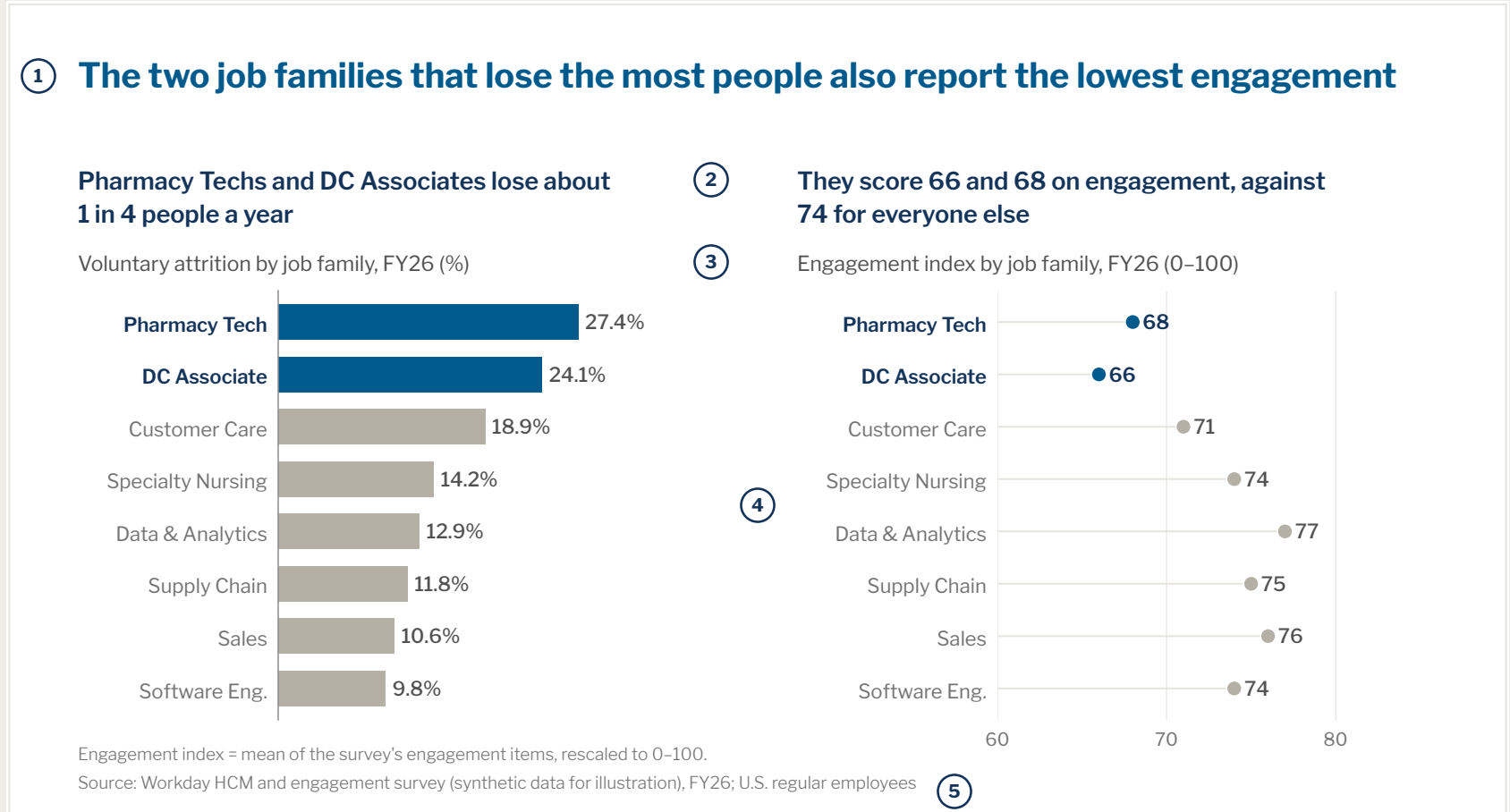
FORMATS

Two charts on a slide

One chart per slide is the default. When the message needs two pieces of evidence, the titles form a hierarchy.

1. **Slide title, Bold**
   The one message both charts support.
2. **Chart headline, SemiBold**
   Each chart's own finding: still an insight, never a label like "Engagement by job family".
3. **Subtitle per chart**
   Metric, unit, scope and period, as always.
4. **Aligned and consistent**
   Headlines on one top line; same rows, same order, same highlight.
5. **One source line**
   Shared at the bottom when the charts share sources.

Two is the limit. A third chart means a second slide.



FORMATS

Flat, not rounded

Rounded pills, soft shadows and gradients date a slide and make it feel less precise. Our look is flat and square.

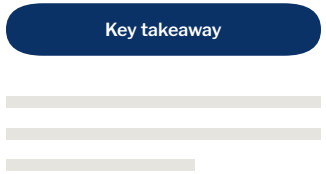
Square corners. Labels, panels and bars have square corners. No pills or capsules.

No shadows or gradients. Separate panels with white space, a sand fill or a thin rule.

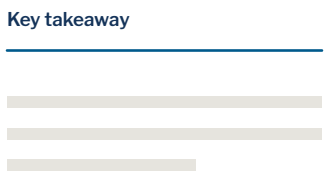
Rules, not boxes. A thin blue rule under a header does the work of a colored capsule.

Words with status dots. A colored dot never stands alone; the word sits beside it.

Labels

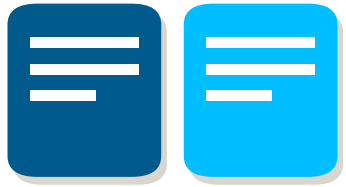


Try to avoid
A rounded pill.



Recommended
Bold label, thin rule.

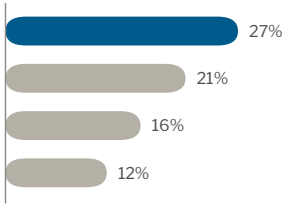
Panels



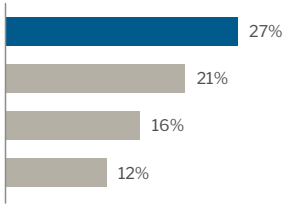
Try to avoid
Rounded, shadowed, bright.

Recommended
Flat sand, a blue rule on top.

Chart marks

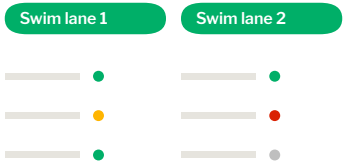


Try to avoid
Rounded bar ends.



Recommended
Square ends, true lengths.

Trackers



Try to avoid
Pills; dots with no words.



Recommended
Plain headers; dot and word.

MAGNITUDE

Bar chart

Magnitude charts compare size: which is bigger (relative), and by how much (absolute).

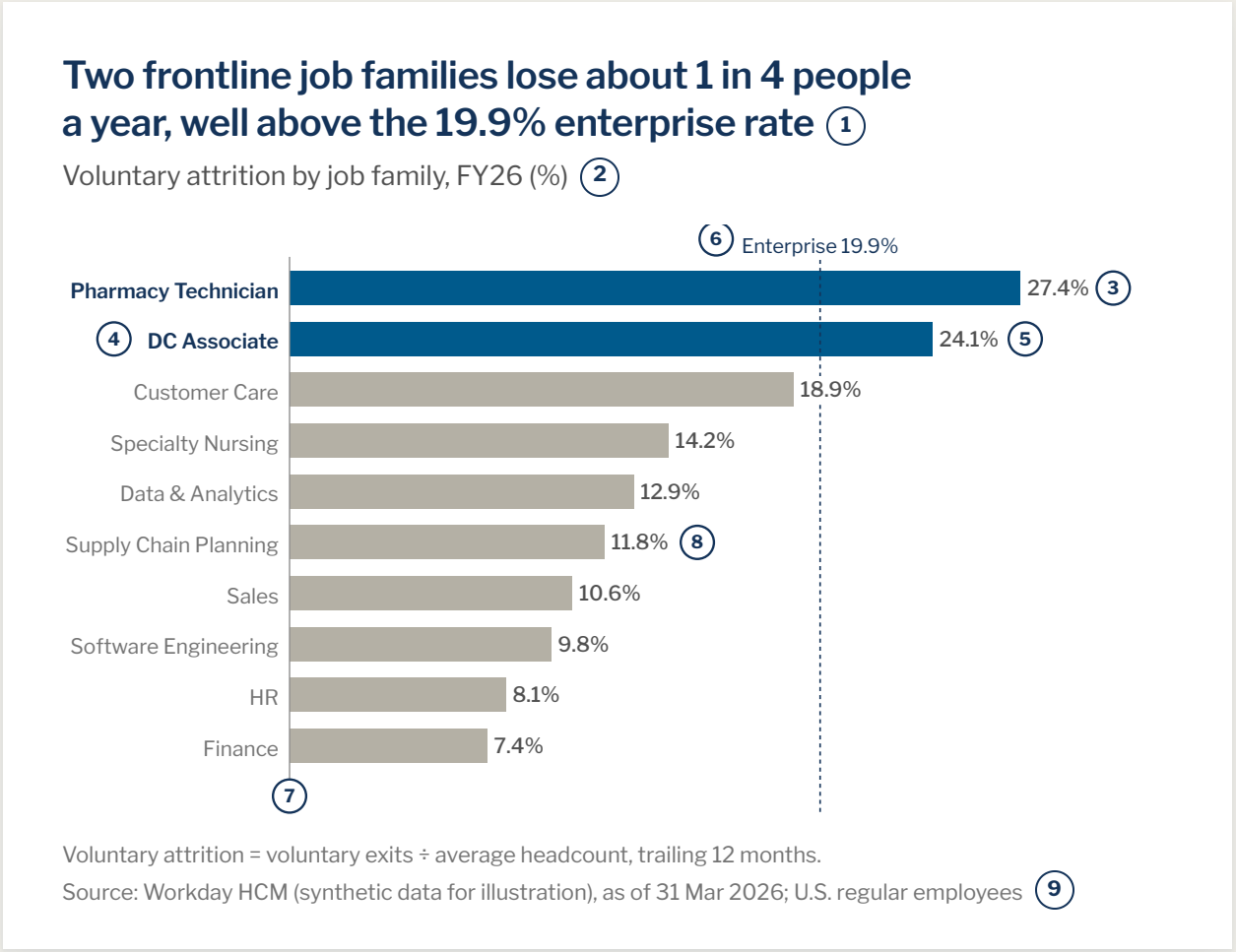
A bar's length is proportional to its value, so bars are the most precise way to compare categories. Sorted horizontal bars are the default for ranking job families, sites or skills: long names stay readable, and the layout works on a phone.

Use columns for a few time periods or short names.

McKesson spec

- Bar about 2× the gap (Excel gap width 50%)
- Sorted by value unless there's a natural order
- Up to about 12 bars; group the tail into "Other"
- No group under 10 people

1. **Insight headline**
   Names the two families and the benchmark they beat.
2. **Subtitle**
   Metric, unit, scope, period.
3. **Focus in blue**
   Only the bars the headline names; two at most.
4. **Echo in the label**
   Focus categories in bold navy.
5. **Values on the bars**
   So the value axis and gridlines go.
6. **Labelled benchmark**
   A thin dashed line, named where it sits.
7. **Zero baseline**
   A thin axis line; bars always start at zero.
8. **Grey context**
   Everything else, sorted by value.
9. **Footnote and source**
   Defines the metric; names system, date, population.



MAGNITUDE

Bar chart best practice

Four rules that fix most bar charts.

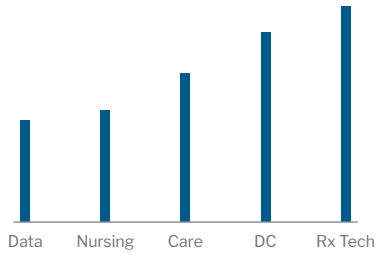
Size of bars. Thin bars pull the eye to the empty space between them. Make each bar about twice as wide as the gap.

Color of bars. A different color per bar distracts and implies categories that don't exist. One color; shade only the highlight.

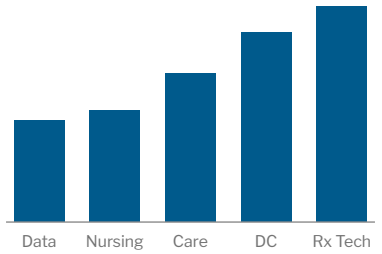
Direct labels. Gridlines and a value axis add noise. Label the bars and delete both.

Zero baseline. Bar length encodes quantity, so never truncate. If the data needs a narrow range, switch to a line or dot plot.

Size of bars

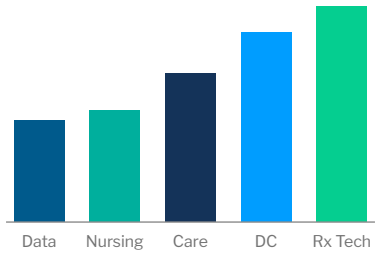


Try to avoid
Stick-thin bars; the gaps dominate.

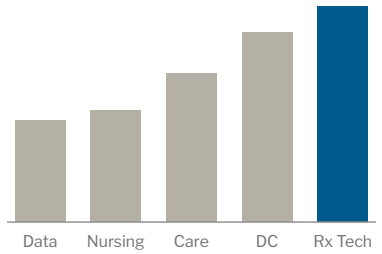


Recommended
Bar about twice the gap.

Color of bars

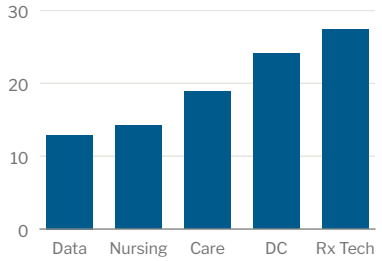


Try to avoid
A color per bar, encoding nothing.

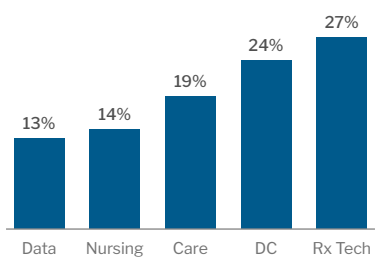


Recommended
Grey bars; blue for the one that matters.

Direct labels

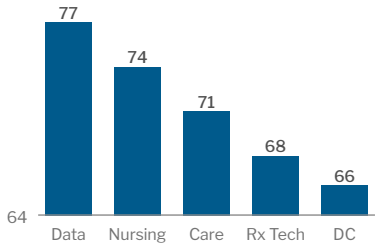


Try to avoid
Gridlines and an axis to read values from.

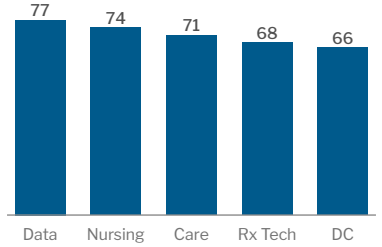


Recommended
Values on the bars; no axis, no grid.

Zero baseline



Try to avoid
Cut at 64, 77 looks 6.5× 66.



Recommended
From zero: 17% higher, as it is.

CHANGE OVER TIME

Line chart

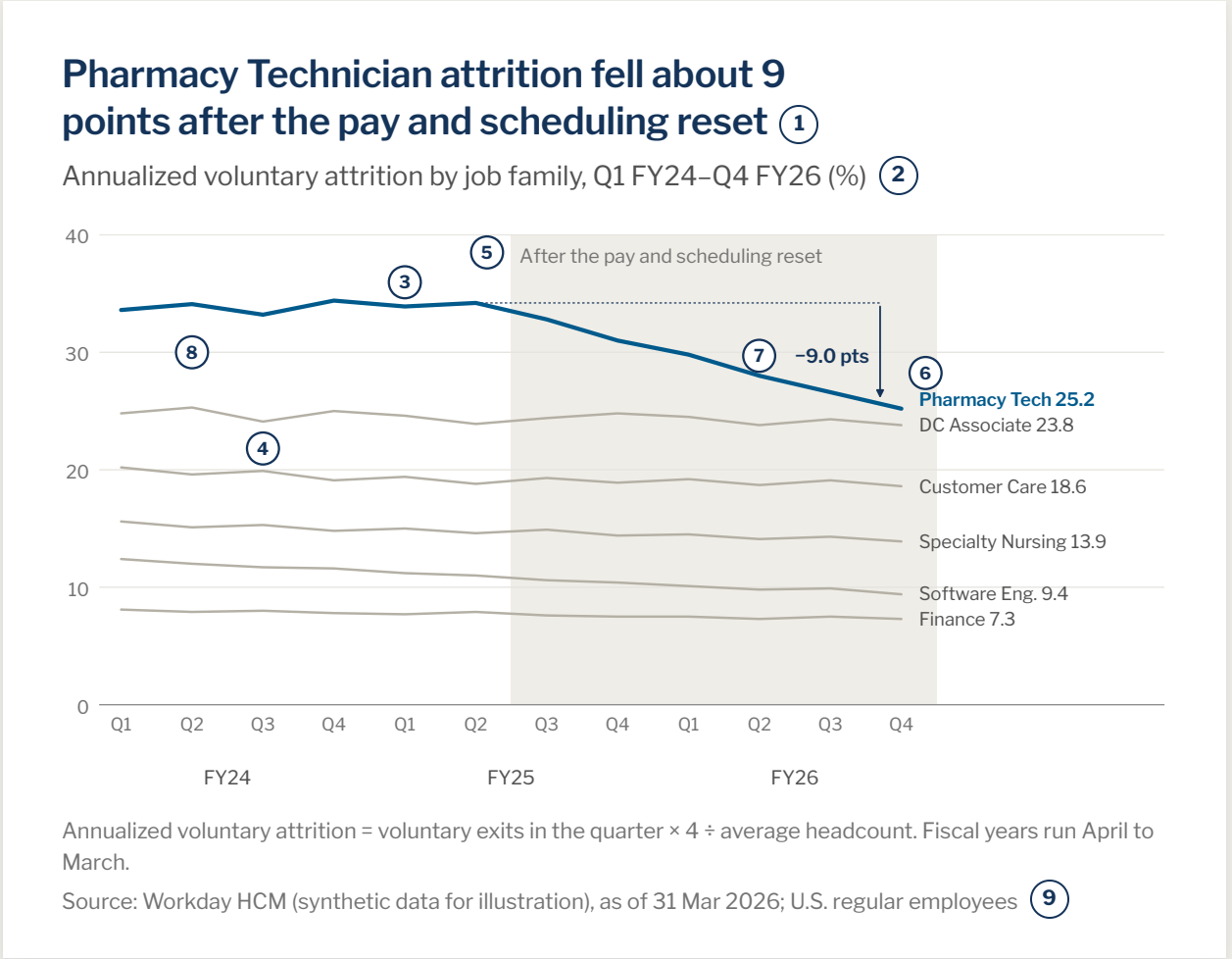
Change-over-time charts show how something moves, and when it changed, at any time scale from weeks to decades.

A line connects values at equal time intervals, so the eye reads the trend rather than the individual amounts. Use it for monthly or quarterly attrition, hiring velocity, or a forecast.

McKesson spec

- The series the headline names in blue; others grey
- Data lines about twice the gridline weight
- Names at the line ends, not in a legend
- Shade the period that explains the change

1. **Insight headline**
   What changed, by how much, and why.
2. **Subtitle**
   Metric, unit, scope and period.
3. **Focus line**
   Blue and heavier than the rest.
4. **Context lines**
   Grey and thinner: there to compare against.
5. **The period that matters**
   Lightly shaded and named.
6. **Direct labels**
   Names and latest values at the line ends, nudged apart.
7. **The change, sized**
   A short callout carries the headline number.
8. **Light scaffolding**
   A few thin gridlines; the unit lives in the subtitle.
9. **Footnote and source**
   Define the metric.



CHEAT SHEET

Cheat sheet

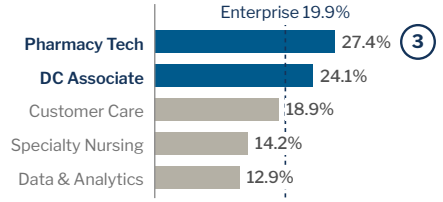
Check before you send

- [ ] The headline is an insight, and every number in it is true
- [ ] Subtitle: metric, unit, scope and period
- [ ] Grey context; at most 2 highlights, matching the headline
- [ ] Values labelled; value axis, gridlines and legend deleted
- [ ] Bars start at zero; no dual axes
- [ ] Sorted by value, or in a natural order
- [ ] True minus (−), explicit + on changes, "pts" for point changes
- [ ] Metric defined; source names system, date and population
- [ ] No group under **10 people**, even in hover text
- [ ] Nothing overlaps or clips; nothing below 9 pt

1 · EVERY CHART, ONE ANATOMY

Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate (1)

Voluntary attrition by job family, FY26 (%) (2)



Source: Workday HCM (synthetic), FY26 (4)

(1) Headline = the insight · (2) Subtitle = the measure · (3) Grey context, blue focus, values on the bars · (4) Source

2 · WRITE THE HEADLINE LAST

Who or what + verb + number + so what. Compute every number from the data.

LABEL	Attrition by job family
VAGUE	Attrition varies across job families
INSIGHT	Two frontline job families lose about 1 in 4 people a year

3 · GREY FIRST, THEN COLOR THE POINT

Context everything else	#B4B0A5
Focus what the headline names	#005A8C
Accent (navy) the one exception	#143359
Light blue a series, in the uniform pattern	#66C4FF
Orange worse only: below target, shrinking	#EF8200
Green better only, with ▲ or words	#00AF68

At most 2 highlights. Orange and green only mean worse and better. Red, amber, green: status only, with a symbol and word.

4 · TYPE (LIBRE FRANKLIN, 16:9 SLIDE)

Slide title	Bold	26 pt
Chart headline	SemiBold	18 pt
Subtitle	Regular	14 pt
Data labels	Medium	12 pt
Axis, categories	Regular	11 pt
Footnote, source	Light	9 pt
Big number	Bold	54 pt

One chart: the slide title is its headline. Two: each its own SemiBold insight headline.

5 · PICK THE CHART FROM THE QUESTION

- How big? Sorted bar
- Made of what? 100% bar, waterfall
- Over time? Line
- Who moved? Slope, dumbbell
- How spread? Strip, histogram
- Related? Scatter
- Flows? Sankey · Where? Map
- One number? Big number

Full version: the chart chooser.

6 · NEVER

- Red vs. green, rainbows, or color as the only cue
- 3D, shadows, gradients, borders
- Truncated bars or dual axes
- A legend where labels fit
- More than 2 charts on a slide
- Pictures of charts: use native, editable charts

TOOLKIT

Set up once

For decks and workbooks, most people need to install nothing: start from the template and copy charts from the starter deck.

Everything named here is in the kit: the release bundle (a zip on SharePoint) or the GitHub repo. More detail and troubleshooting:

- docs/office-setup.md
- docs/powerbi-setup.md
- AGENTS.md (Python, HTML, R)

Check the font	In PowerPoint, type **Libre Franklin** in the font box. If it's listed, you're set. If not, open the kit's **fonts** folder, select every file, right-click › **Install** (or ask IT).
PowerPoint	**New deck:** File › New › the **McKesson Standard Template**. **Charts:** open the kit's **starter deck**, copy a chart slide into your deck, then right-click the chart › **Edit Data** and paste your numbers. Colors, fonts and layout come with the slide.
Excel	Open the kit's **chart workbook**: replace a sheet's data, or copy its chart into your file. The **Palette** sheet lists every color with the hex to type into More Colors. Want the colors in your own workbooks? The optional theme files are in office-setup.
Power BI	**View › Themes › Browse for themes** and pick `dist/powerbi/mckesson-dataviz-theme.json`. Viewers without Libre Franklin see Segoe UI, so leave room in labels.

Python	`git clone https://github.com/amccown-mck/Data-Viz-Guide  # then, in it:`
	`pip install -e "./python[examples]"  # editable, so the fonts are found`
	`import mckviz as mv`
	`mv.use()  # fonts, matplotlib style, Plotly template "mck"`
	`C = mv.colors  # C.focus, C.context, C.accent, C.negative ...`

HTML	Link `dist/web/mck-viz.css` for the CSS variables and chart-card styles; add `mck-plotly-template.js` for Plotly or use `vega-lite-config.json` as the Vega-Lite `config`. Start from `templates/html/chart-card.html`.
R	`source("dist/r/theme_mck.R")`, then `ff <- mck_font()` and `theme_mck(base_family = ff)`. Render with `ragg`.